# DATABASE SYSTEM ARCHITECTURE

# Database Architecture

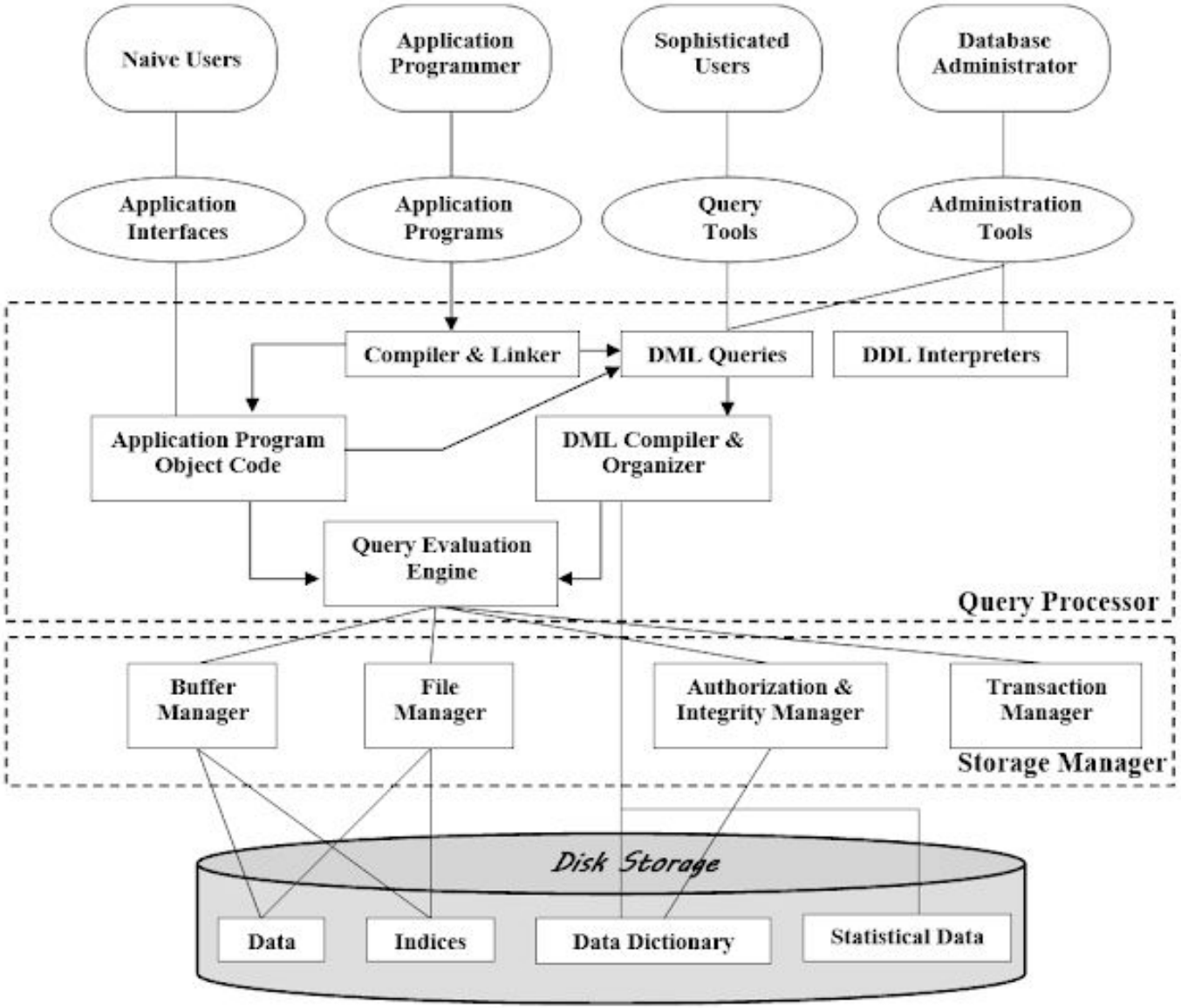



*Figure: System Architecture*

# Components of Database Architecture

1. Database Users
2. Database Administrator
3. Query Processor
4. Storage Manager
5. Disk Storage

# 1. Database Users

- Users are differentiated by the way they expect to interact with the system:
- **Naïve users :**
  - Naive users are unsophisticated users **who interact with the system by invoking one of the application programs that have been written previously**.
    - For example, a bank teller who needs to transfer $50 from account A to account B invokes a program called transfer. This program asks the teller for the amount of money to be transferred, the account from which the money is to be transferred, and the account to which the money is to be transferred.

- **Application programmers:**
  - Application programmers are **computer professionals who write application programs.**
  - Application programmers can choose from many tools to develop user interfaces.
  - **Rapid application development (RAD) tools** are tools that enable an application programmer **to construct forms and reports without writing a program**

# 1. Database Users

- Users are differentiated by the way they expect to interact with the system:
- **Sophisticated users:**
  - Sophisticated users interact with the system **without writing programs**. Instead, they form their requests in a **database query language**.
  - They submit each such query to a query processor, whose function is to break down DML statements into instructions that the storage manager understands.

- **Specialized users :**
  - Specialized users are sophisticated users who write **specialized database applications that do not fit into the traditional data-processing framework**.
  - Among these applications are **computer-aided design systems**, knowledge base and expert systems, systems that store data with complex data types (for example, **graphics data and audio data), and environment-modeling systems.**

# 2. Database Administrator

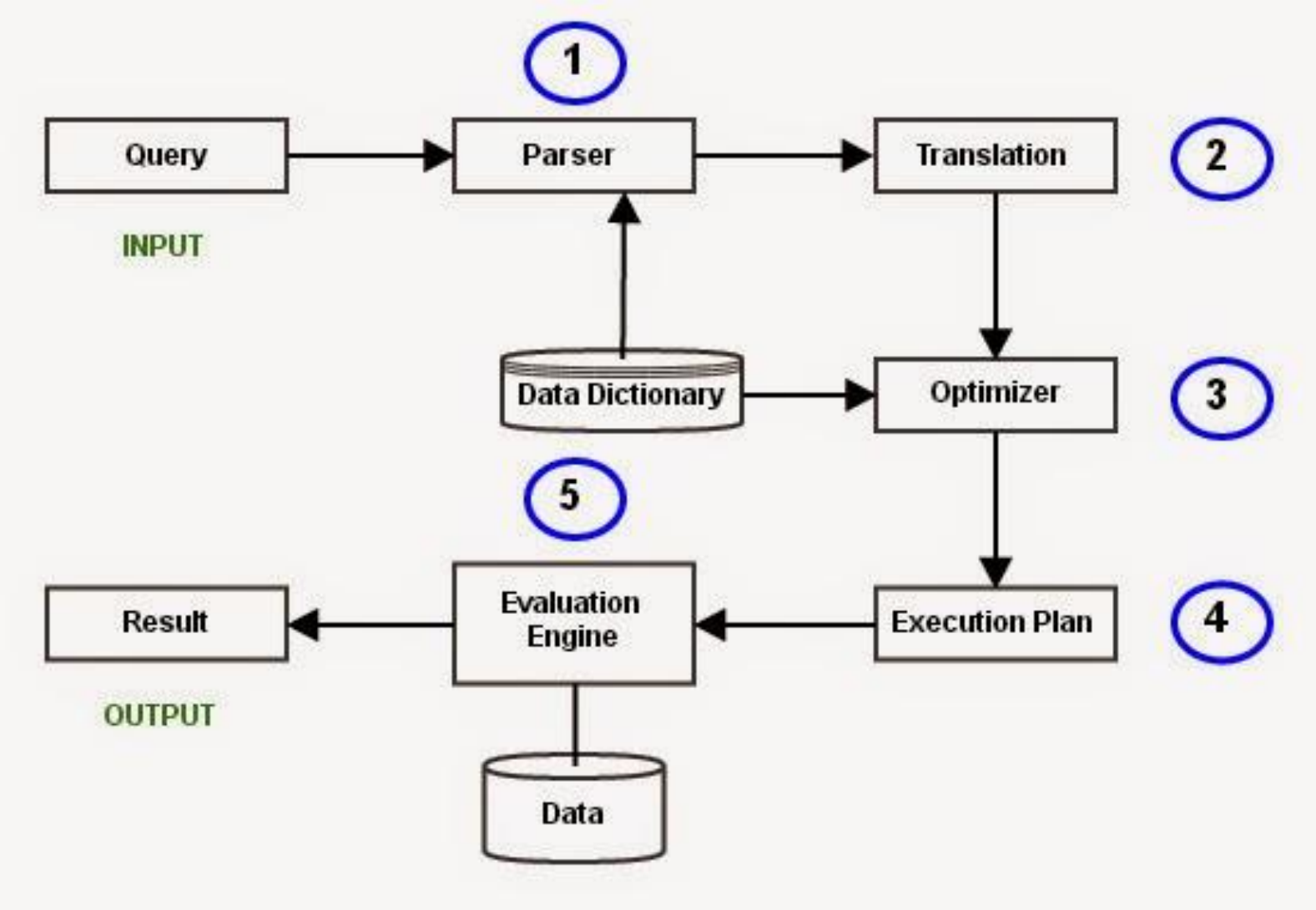
- **Coordinates** all the activities of the database system. The database administrator has a good understanding of the enterprise's information resources and needs.
- Database administrator's duties include:
  - **Schema definition:** The DBA creates the original database schema by executing a set of data definition statements in the DDL.
  - **Storage structure and access method definition.**
  - **Schema and physical organization modification:** The DBA carries out changes to the schema and physical organization to reflect the changing needs of the organization, or to alter the physical organization to improve performance.
  - **Granting user authority to access the database:** By granting different types of authorization, the database administrator can regulate which parts of the database various users can access.
  - **Specifying integrity constraints.**
  - **Monitoring performance and responding to changes in requirements.**

# 3. Query Processor

- The query processor will **accept query from user and solves it by accessing the database**.

- **Parts of Query processor:**
  - **DDL interpreter**
    - This will **interpret DDL statements and fetch the definitions in the data dictionary**.
  - **DML compiler**
    - This will **translate DML statements in a query language into low level instructions that the query evaluation engine understands**.
    - A query can usually be translated into any of a number of alternative evaluation plans for same query result DML compiler will **select best plan for query optimization**.
  - **Query evaluation engine**
    - This engine will execute **low-level instructions generated by the DML** compiler on DBMS.

# Query Parsing



# 4. Storage Manager

- A storage manager is a program module which acts like **interface between the data stored in a database and the application programs and queries submitted to the system**.
- The storage manager is responsible for **storing, retrieving and updating data** in the database.
- **The storage manager components include:**
  - **Authorization Manager:** It ensures role-based access control, i.e,. checks whether the particular person is privileged to perform the requested operation or not.
  - **Integrity Manager:** It checks the integrity constraints when the database is modified.
  - **Transaction Manager:** It controls concurrent access by performing the operations in a scheduled way that it receives the transaction. Thus, it ensures that the database remains in the consistent state before and after the execution of a transaction.
  - **File Manager:** It manages the file space and the data structure used to represent information in the database.
  - **Buffer Manager:** It is responsible for cache memory and the transfer of data between the secondary storage and main memory.

# 4. Storage Manager (Cont.)

- The storage manager implements several data structures as part of the physical system implementation:
  - Data files -- store the database itself
  - Data dictionary -- stores metadata about the structure of the database, in particular the schema of the database.
  - Indices -- can provide fast access to data items. A database index provides pointers to those data items that hold a particular value.

# 5. Disk Storage

- The storage manager implements several data structures as part of the physical system implementation:

  - **Data files** -- store the **database itself**

  - **Data dictionary** -- **stores metadata** about the structure of the database, in particular the schema of the database

  - **Indices** -- can provide **fast access** to data items. A database index **provides pointers** to those data items that hold a particular value

# DATABASE ARCHITECTURE - TYPES

# Database Architectures - Types

The architecture of a database system is greatly influenced by the underlying computer system on which the database runs. Based on the underlying systems, the database architecture can be:

1. Centralized databases
    - One to a few cores, shared memory
2. Client-server,
    - One server machine executes work on behalf of multiple client machines.
3. Parallel databases
    - Many core shared memory
    - Shared disk
    - Shared nothing
4. Distributed databases
    - Geographical distribution
    - Schema/data heterogeneity

## Centralized Database Architecture

A centralized database is a database that is located, stored, and maintained in a single location. This location is most often a central computer or database system, for example a desktop or server CPU, or a mainframe computer.

It is maintained and modified from that location only and usually accessed using an internet connection such as a LAN or WAN. The centralized database is used by organizations such as colleges, companies, banks etc.

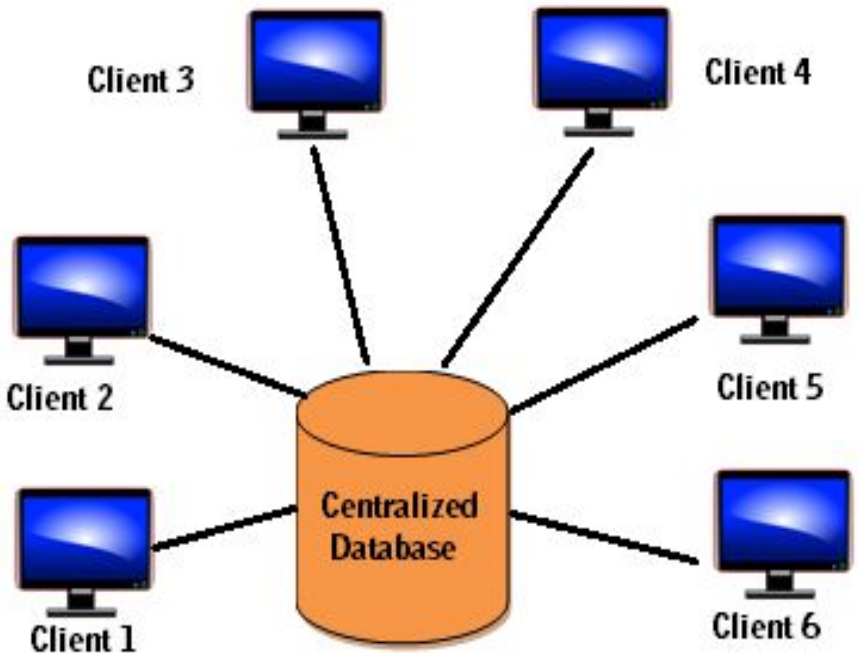



*Fig: Centralized database system*

## Client/Server Database Architecture

The client/server architecture is based on the hardware and software components that interact to form a system. The system includes three main components: **Clients**, **Servers** and **Communication Middleware**.

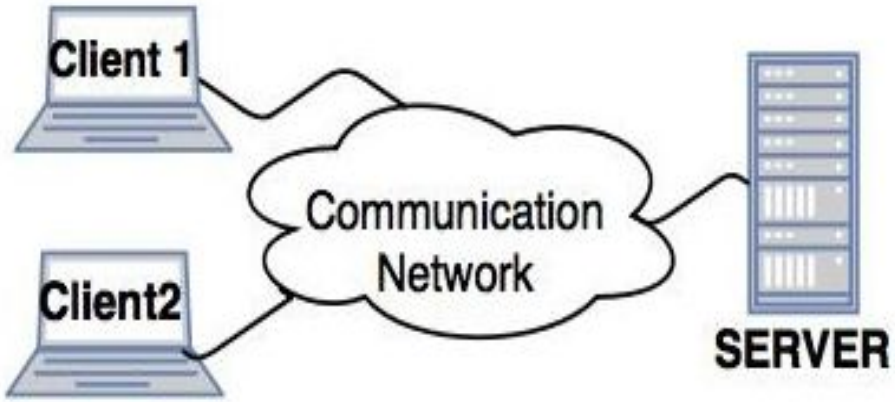



***Fig: Client/server system***

**Client:** The client is any computer process that requests service from the server.

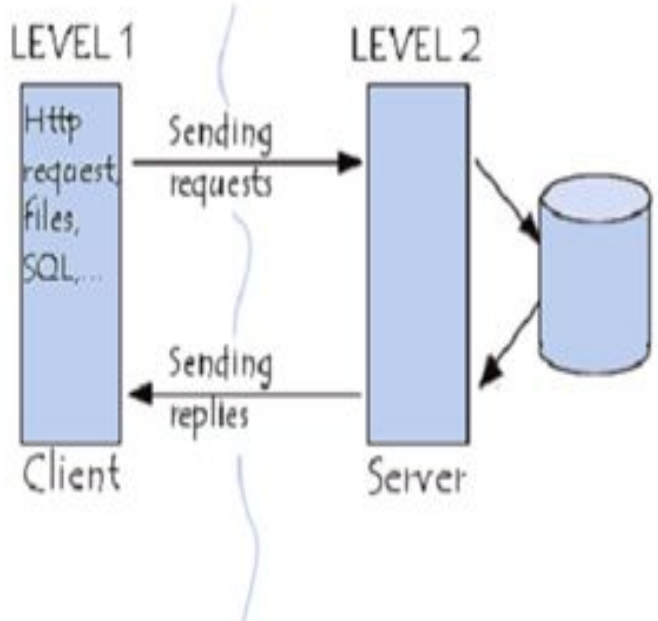
**Server:** The server is any computer process providing services to the clients.

**Communication Middleware:** The communication middleware is any computer process through which clients and servers communicate and is also known as **communication layer**.

There are basically **two-types of client/server architectures**:

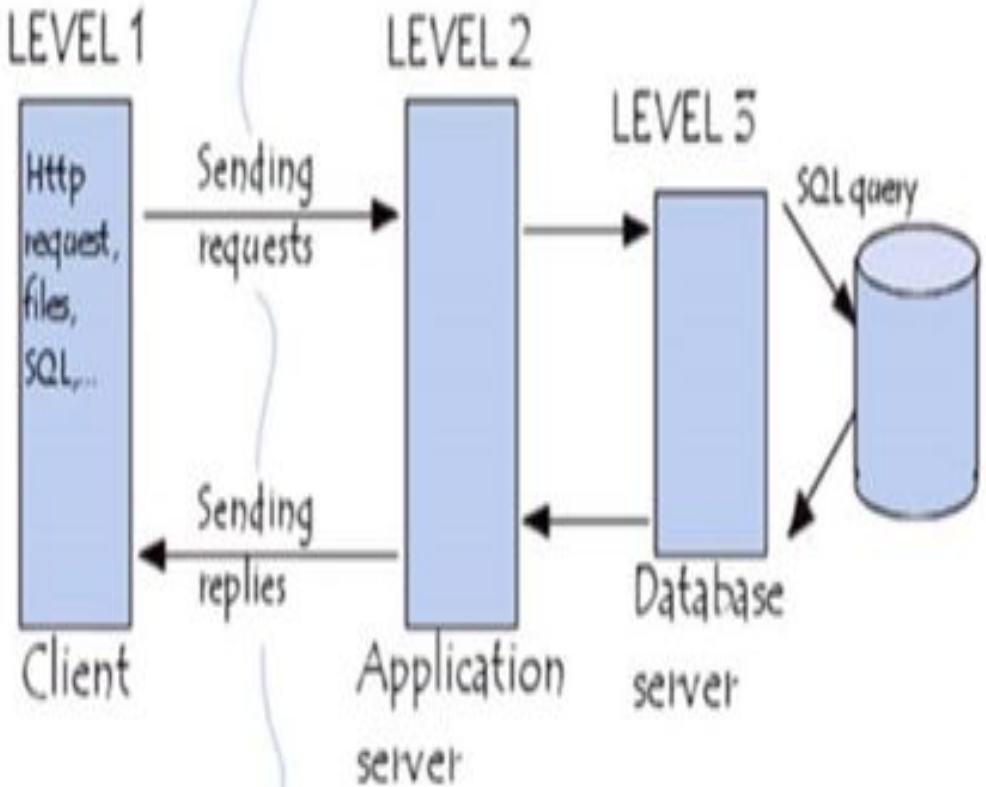
➢ ***Two-tier Architecture:***

In his architecture, the user interface and application programs are placed on the client side and database system on the server side. The application programs that resides at the client side invoke the DBMS at the server side.



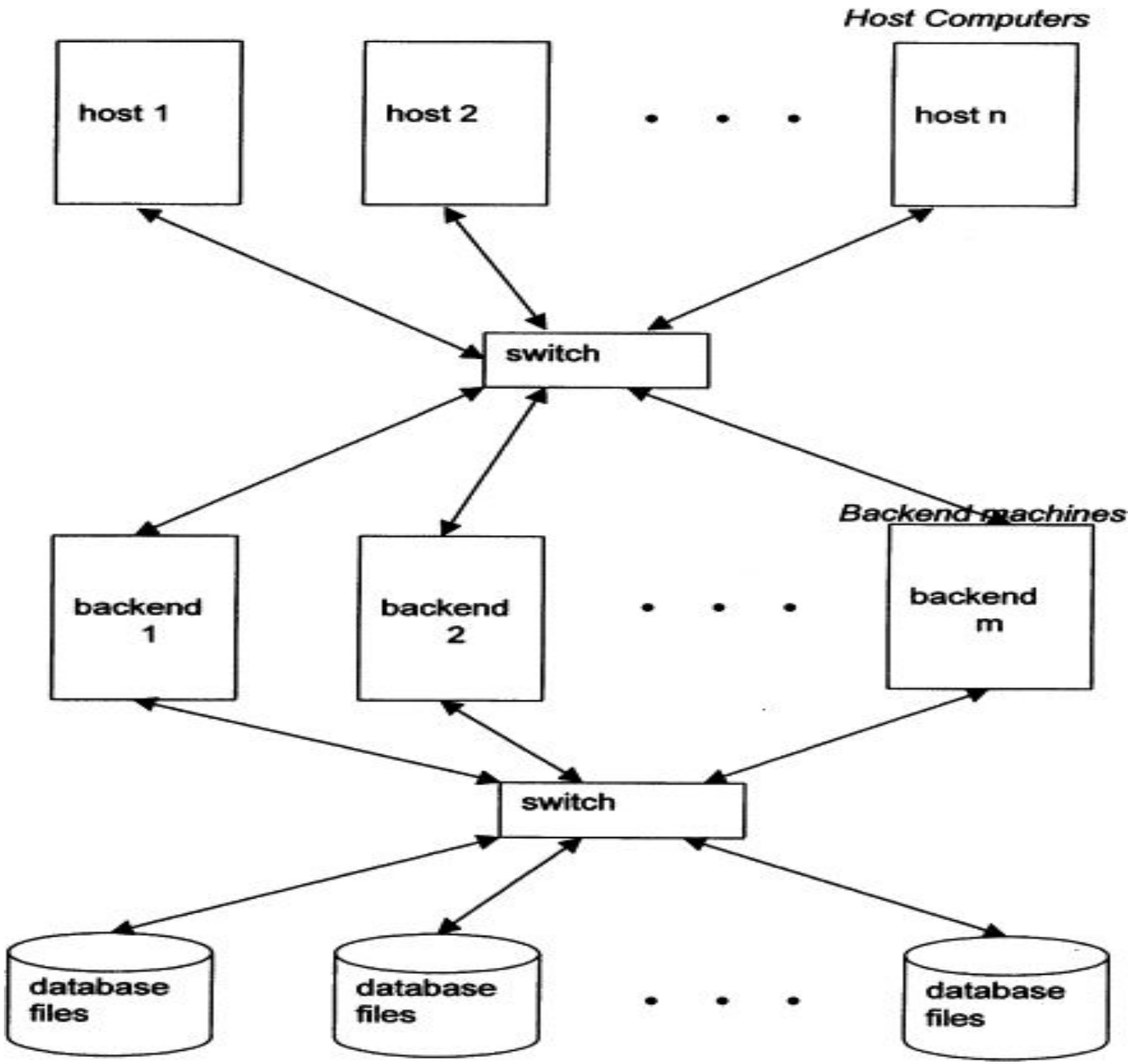
➢ ***Three-tier Architecture:***

This architecture adds application server between the client and database server. The client communicates with the application server, which in turn communicates with the database server. The application server stores the business rule (procedures and constraints) used for accessing data from database server.



**Parallel Architecture:**

In a parallel database machine environment, instead of a single backend, there are multiple backend machines connected either to a single host or to multiple hosts. The backends control multiple disk units containing the database.



## Distributed Database Architecture

A distributed database is a set of logically interrelated database that are stored on computers at several geographically different sites and are linked by means of a computer network. These interrelated database work together to perform certain specific tasks.
Distributed computer system work by splitting a large task into a number of smaller ones that can then be solved in a coordinated fashion. Each processing element can be managed independently and can be developed its own applications.

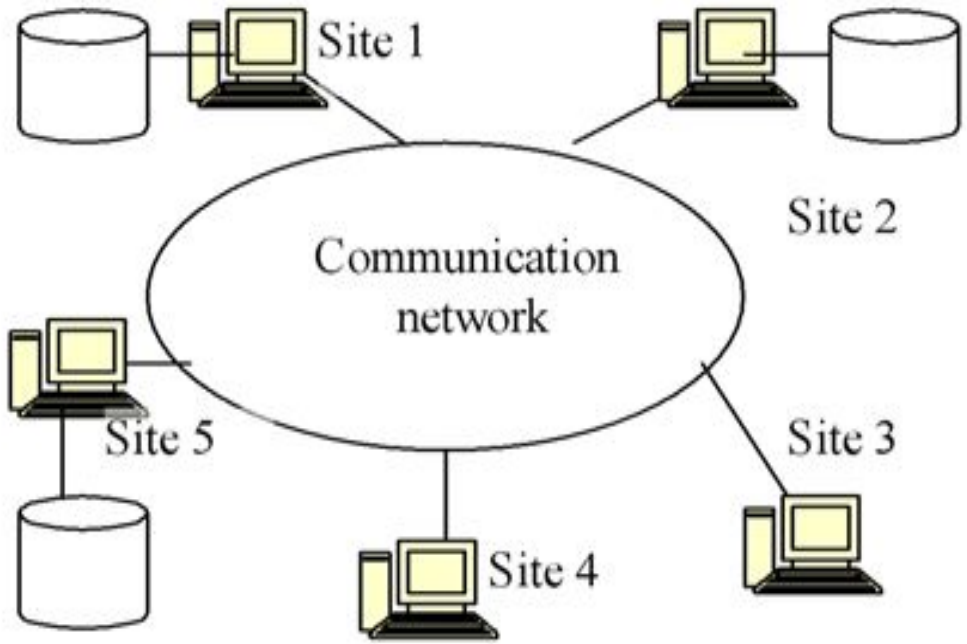



*Fig: Distributed database system*